

Characterizing Single-GPU Decoder-Only Transformer Inference: Precision, Compilation, Batching, and Model Size on an RTX 5070

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Abstract

We present a controlled single-GPU study of decoder-only transformer inference using a small open causal language model. On an NVIDIA GeForce RTX 5070, we measure synchronized end-to-end `generate()` latency and tokens-per-second across a 32-cell matrix spanning data type (FP16, BF16), batch size (1, 4), prompt length (64, 256), generation length (32, 128), and execution mode (eager vs. `torch.compile`). All cells completed without out-of-memory failures. BF16 reduced end-to-end latency relative to FP16 in every paired comparison (median BF16/FP16 latency ratio 0.90). Increasing batch size from 1 to 4 raised throughput by roughly $4\times$ while keeping wall-clock latency in a similar range. A follow-up batch-size sweep (1 to 256) locates the latency-throughput knee: latency stays near 2.3 s up to batch 32 while throughput rises about $31\times$, after which latency grows roughly linearly with batch size and throughput rises only from 2,222 tok/s at batch 64 to 2,467 tok/s at batch 256. Repeating the sweep on 360M- and 1.7B-parameter models from the same family shows the knee moving to smaller batches as the model grows (batch 16 for 360M; no flat region for 1.7B), and shows that per-token latency at batch 1 does not follow parameter count: the 1.7B model decodes about 40% faster per step than the 135M model. For the 1.7B model, the default growing KV cache fragments PyTorch’s allocator enough that batch 64 reserves 24 GB for 8 GB of live data and silently spills into system memory; a preallocated static cache avoids this and reaches 2,044 tok/s. Generation length dominated end-to-end time, while tokens-per-second stayed comparatively flat, consistent with decode-bound behavior under this harness. `torch.compile` yielded a median eager/compile latency ratio of 1.02 (range 0.92–1.22), indicating little systematic gain for this model and measurement window. We release the harness, configurations, and aggregated results at <https://github.com/benkarlsberg/GPU-Optimized-Transformer-Inference>.

1 Introduction

Autoregressive transformer inference [1] is commonly deployed with mixed precision, compiler stacks, and request batching. Practitioners must choose among these options without assuming that every optimization helps every workload. This paper reports a small, reproducible characterization on one consumer GPU using the open-source `goti` harness accompanying this work. Our goal is empirical clarity on a fixed model and hardware point—not a claim of state-of-the-art serving performance.

Serving systems batch requests on the premise that small-batch decoding leaves the GPU underused [20, 12, 14]. How far batching stays free before latency rises depends on the model and the device, and is rarely reported for small models on consumer GPUs through the standard Hugging Face `generate()` path. Our contributions are (i) a 32-cell factorial measurement of precision, batch size, prompt length, generation length, and `torch.compile` for a 135M-parameter model on an RTX 5070; (ii) a batch-size sweep from 1 to 256 that locates the latency-throughput knee, with an explicit account of an excluded noisy pass; (iii) a repetition of the sweep across three model sizes and two KV-cache implementations, including a memory-fragmentation failure mode that is invisible to allocated-memory metrics; and (iv) an open harness that records the configuration, environment, and allocated, reserved, and driver-reported memory of every run.

2 Related work

Serving systems and batching. Orca [20] introduced iteration-level scheduling so that new requests can join a running batch, and vLLM [12] stores the KV cache in fixed-size pages so that more sequences fit in memory at once. Sarathi-Serve [3] studies the throughput–latency tradeoff created when prefill and decode share a batch. Pope et al. [14] model inference cost as a function of batch size, memory bandwidth, and partitioning on TPUs, and DeepSpeed-Inference [5] and FlexGen [17] target models too large for a single device or its memory. These systems motivate batching; our sweep measures where its benefit ends for a small model, using static batches rather than a serving engine.

Kernels and attention. FlashAttention [8, 7] reduces attention memory traffic by tiling the computation, and PyTorch’s scaled-dot-product attention dispatches to fused kernels of this kind. Multi-query attention [16] shrinks the KV cache read at every decode step. Our harness uses the stock attention path and does not modify kernels.

Precision, quantization, and compilation. FP16 mixed precision [13] and BF16 [11] were introduced mainly for training speed and numerical range; their relative latency at inference on a given GPU is an empirical question that we measure directly. Weight quantization such as LLM.int8() [9] and GPTQ [10] further reduces memory traffic; we leave it to future work. `torch.compile` [6] captures Python programs as graphs (TorchDynamo) and generates fused kernels (TorchInductor); we evaluate its default mode.

Benchmarking methodology. MLPerf Inference [15] standardizes inference measurement across systems, and the roofline model [18] relates achieved performance to compute and bandwidth limits. Our study is narrower: a controlled factor sweep on one consumer GPU, intended as a reproducible reference point rather than a benchmark suite.

3 Method

3.1 Software stack

Experiments use PyTorch [2] with CUDA 12.8 bindings (`torch 2.11.0+cu128`), Hugging Face Transformers [19] for model loading, and scaled-dot-product attention via PyTorch SDPA backends. The workload is `greedy model.generate()` with KV caching enabled. Synthetic token ids form prompts so logging of decoded text cannot contaminate timed regions.

3.2 Metrics and timing protocol

For each configuration we run a warmup phase followed by measured iterations. Each iteration records GPU-synchronized end-to-end latency of `generate()`, peak CUDA memory allocated, and tokens-per-second derived from generated token count and wall time. We report distributional summaries (mean, selected percentiles) rather than a single timing. Cold-start model load is outside the measured loop; `torch.compile` configurations still include steady measured iterations after warmup, but compilation cost can affect short windows and is treated as part of the system under test.

3.3 Experimental matrix

Model: HuggingFaceTB/SmolLM-135M [4], a 135M-parameter decoder-only language model. Factors: `dtype` $\in \{\text{fp16}, \text{bf16}\}$, batch size $\in \{1, 4\}$, prompt length $\in \{64, 256\}$, max new tokens $\in \{32, 128\}$, compile

$\in \{\text{false}, \text{true}\}$. This yields 32 cells. Hardware: a single desktop workstation with an NVIDIA GeForce RTX 5070 (≈ 12 GB, NVIDIA 610-series driver) running Windows 11.

4 Results

All 32 cells completed with no OOM events. The lowest mean end-to-end latency was 586 ms (BF16, batch 4, prompt 256, gen 32, compile enabled; ≈ 218 tok/s). The highest was 2855 ms (FP16, batch 4, prompt 64, gen 128, eager; ≈ 180 tok/s).

4.1 Precision

Figure 1 compares FP16 and BF16 latency for every matched non-dtype factor. BF16 was faster in all 16 pairs. The median BF16/FP16 latency ratio was 0.895 (mean 0.890).

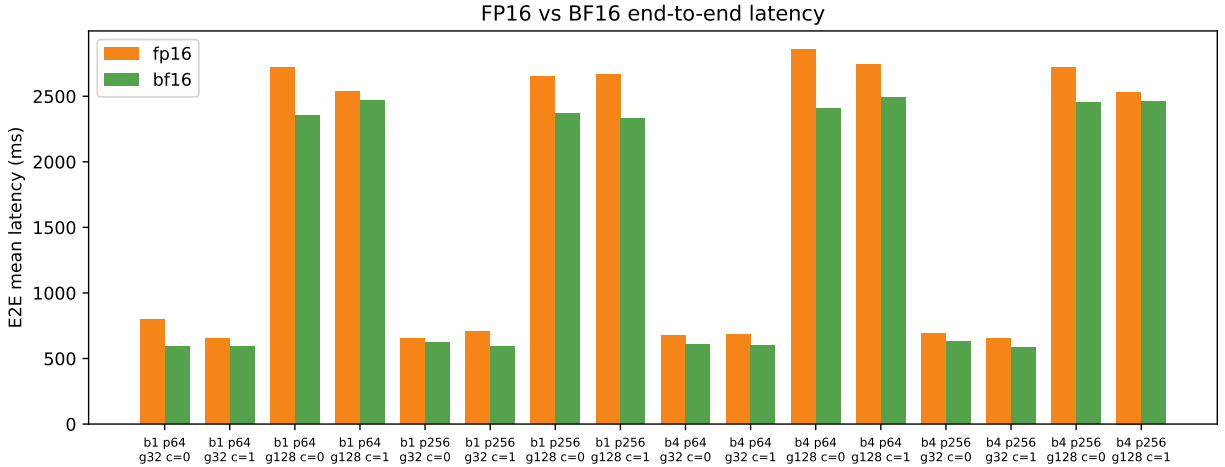


Figure 1: End-to-end mean latency for FP16 vs. BF16 across matched matrix cells.

4.2 Batching and throughput

Figure 2 shows tokens-per-second versus batch size for FP16 eager runs. Moving from batch 1 to batch 4 increased throughput from 41–49 tok/s to 180–189 tok/s. End-to-end latency did not rise with batch size in this range: going from batch 1 to batch 4 raised throughput without a latency penalty. With only two batch sizes, however, the matrix cannot show where that stops, so we ran a dedicated sweep (Section 4.3).

4.3 Batch-size sweep: the latency–throughput tradeoff

To locate where batching stops being free, we swept batch size over $\{1, 2, 4, \dots, 256\}$ with all other factors fixed (BF16, eager, prompt 256, 128 new tokens; 2 warmup and 5 measured iterations per cell). We ran the sweep three times. The first pass was inconsistent with the matrix (batch 1 took 3.4 s versus 2.37 s in the matrix, and batch 256 was faster than batch 128), which we attribute to host-side variability on the workstation during that pass; it is excluded from the aggregate but kept in the released analysis file. The second pass ran batch sizes in descending order and the third in ascending order; they agreed within 2% at every batch size, and Table 1 and Figure 3 report their mean. No cell ran out of memory; peak allocated memory was 4.0 GB at batch 256.

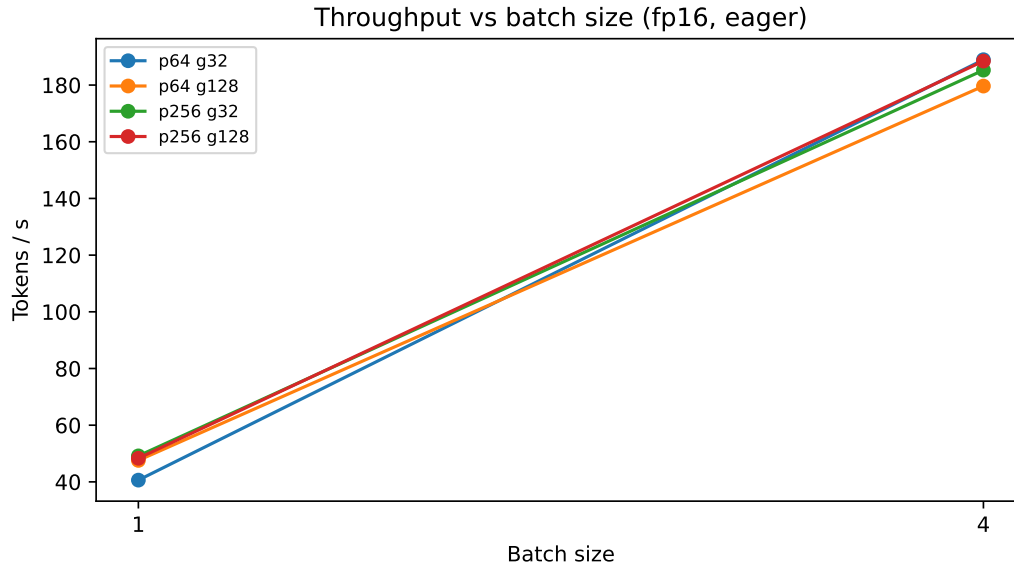


Figure 2: Throughput versus batch size (FP16, eager).

Batch	Latency (ms)	Throughput (tok/s)	Peak memory (MB)
1	2303	56	283
2	2314	111	299
4	2354	218	328
8	2335	438	386
16	2317	884	502
32	2336	1754	733
64	3687	2222	1197
128	6827	2401	2126
256	13286	2467	3980

Table 1: Batch-size sweep (BF16, eager, prompt 256, 128 new tokens), mean of two passes.

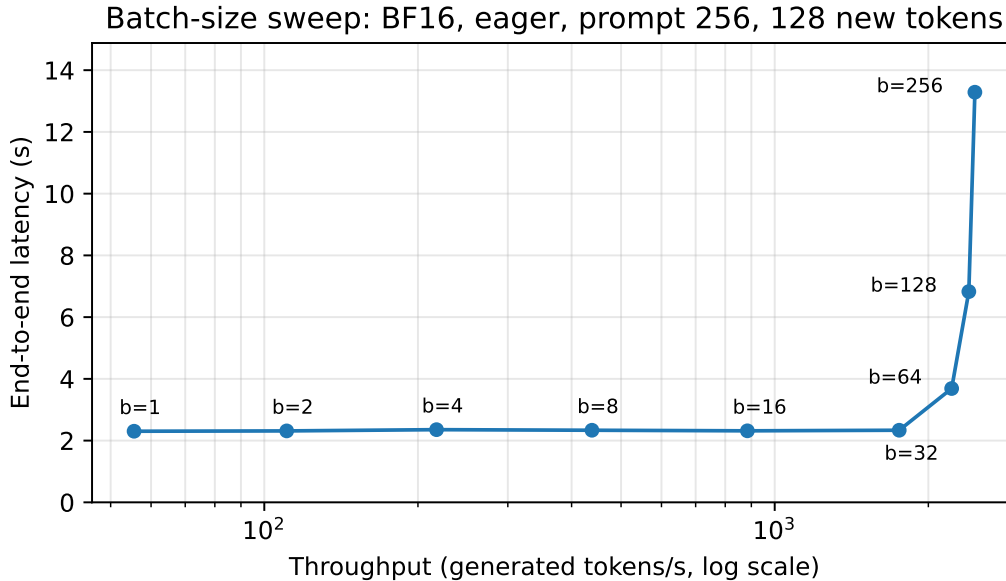


Figure 3: Latency versus throughput across the batch-size sweep. Each point is one batch size; the curve is flat up to batch 32 and then turns sharply upward.

Two regimes are visible. From batch 1 to batch 32, end-to-end latency stays between 2.30 and 2.35 s (about 18 ms per decode step) while throughput rises from 56 to 1,754 tok/s. In this regime a decode step for this 135M-parameter model does not occupy the GPU, so per-step time is set by fixed costs that do not grow with batch size, and additional sequences are nearly free. Beyond batch 32 the GPU becomes the bottleneck: latency grows roughly in proportion to batch size (3.7 s at 64, 6.8 s at 128, 13.3 s at 256) while throughput rises only from 2,222 tok/s at batch 64 to 2,467 tok/s at batch 256. For a latency target, the useful operating point on this hardware is therefore around batch 32; batch 64 buys about 27% more throughput at a 58% latency cost, and larger batches add little throughput. We did not profile the decode step, so the attribution of the flat regime to fixed per-step costs is an interpretation of the curve shape rather than a measured breakdown.

4.4 Model size and KV-cache allocation

To test whether the knee is a property of the GPU or of the model, we repeated the sweep with two larger members of the same family, SmoILM-360M and SmoILM-1.7B [4], under the same settings. Table 2 lists their shapes; the two smaller models are deep and narrow and use grouped-query attention, while the 1.7B model has fewer, wider layers and full multi-head attention. Each configuration again ran twice, once in descending and once in ascending batch order; the two passes agreed within 1.5% at every valid cell. These runs were made in a later session, after the workstation’s power plan had been changed, so we also re-ran the 135M sweep in that session. It reproduced the earlier sweep within 1.1% at every batch size up to 128 and was 4.5% faster at batch 256; Table 3 uses the re-run so that all three models share one session. Besides allocated memory, the harness now records PyTorch’s reserved memory, the driver-reported memory in use, and the Windows shared-GPU-memory counter, which reveals allocations that the driver has placed in system RAM instead of failing.

The knee moves left as the model grows. Latency for the 360M model stays between 2.43 and 2.48 s through batch 16 and then rises to 3.50 s at batch 32, one doubling earlier than the 135M model. The 1.7B

Model	Layers	Hidden	Query heads	KV heads	BF16 weights
SmolLM-135M	30	576	9	3	0.27 GB
SmolLM-360M	32	960	15	5	0.72 GB
SmolLM-1.7B	24	2048	32	32	3.4 GB

Table 2: Architectures of the three models (from their published configurations).

Batch	135M		360M		1.7B dyn.		1.7B static	
	ms	tok/s	ms	tok/s	ms	tok/s	ms	tok/s
1	2308	55	2457	52	1379	93	1706	75
2	2332	110	2437	105	1514	169	1732	148
4	2365	216	2430	211	1614	317	1748	293
8	2330	440	2451	418	1801	569	1848	554
16	2341	875	2481	826	2233	917	2154	951
32	2333	1756	3504	1169	3232	1268	2751	1489
64	3680	2226	6301	1300	spilled [†]		4008	2044
128	6754	2426	11544	1419	–	–	–	–
256	12684	2583	22282	1471	–	–	–	–

Table 3: End-to-end latency and throughput across model sizes; “dyn.” is the default growing KV cache (BF16, eager, prompt 256, 128 new tokens; mean of two passes). [†]About 13 GB was placed in system memory and latency rose to 35 s; the cell is excluded. Dashes mark batch sizes that were not run because they cannot fit in 12 GB.

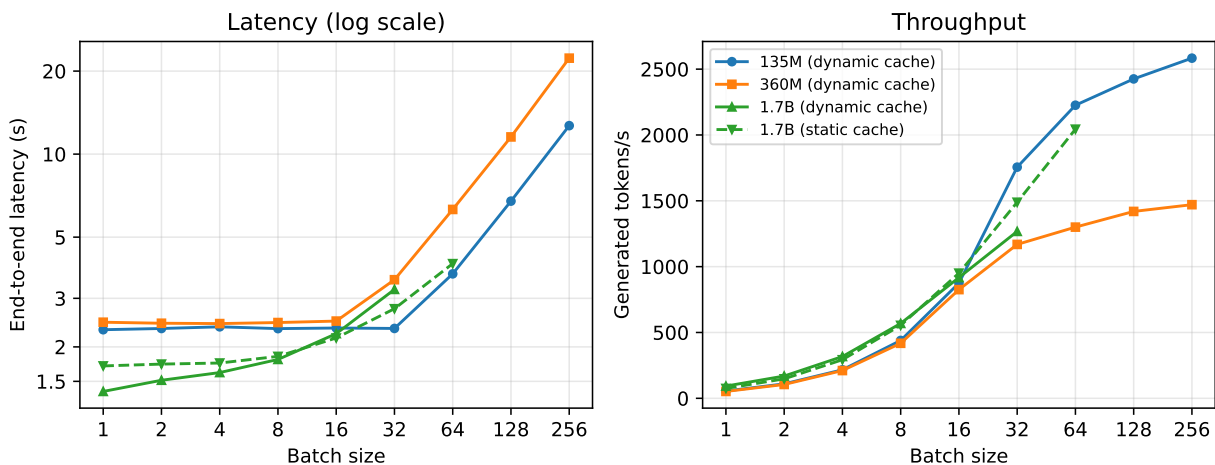


Figure 4: Latency (left, log scale) and throughput (right) versus batch size for three model sizes. The dashed line is the 1.7B model with a static KV cache; the spilled 1.7B dynamic-cache cell at batch 64 is omitted.

model has no flat region: relative to batch 1, latency is 17% higher at batch 4, 31% higher at batch 8, and 62% higher at batch 16. The regime in which extra sequences are nearly free therefore shrinks with model size and has disappeared by 1.7B parameters on this GPU.

Per-step latency does not follow parameter count. At batch 1 the three models take 18.0, 19.2, and 10.8 ms per decode step respectively. The 360M model has $2.7\times$ the parameters of the 135M model but is only 6% slower, and the 1.7B model, with $13\times$ the parameters, is 40% faster. Per layer, the two smaller models take about 0.60 ms and the 1.7B model about 0.45 ms. Even for the 1.7B model, reading its 3.4 GB of weights at the card’s rated 672 GB/s memory bandwidth would take about 5 ms, roughly half of the measured step. These numbers are consistent with small-batch decoding being limited by a per-layer cost that depends on layer structure rather than on weight bytes; the grouped-query attention path in the smaller models may contribute additional operations per layer. We have not profiled the step, so this remains an interpretation. The same pattern holds for throughput: the 360M model saturates at 1,471 tok/s, below both the 135M model (2,583) and the 1.7B model with a static cache (2,044 at batch 64).

KV-cache allocation and silent spilling. The default cache in Transformers grows by one position per decode step. For the two smaller models, whose grouped-query caches are small, PyTorch’s reserved memory never exceeds allocated memory by more than 18% (at most about 0.2 GB). For the 1.7B model, whose full multi-head cache is about 75 MB per sequence at these lengths, the ratio of reserved to allocated memory grows from 1.06 at batch 1 to 1.29 at batch 16, 1.72 at batch 32 (9.7 versus 5.6 GB), and 3.06 at batch 64 (24.4 versus 8.0 GB). On Windows the driver satisfied the excess from system memory instead of raising an out-of-memory error: about 13 GB was placed in shared memory and the batch-64 cell took 35.1 s. Allocated-memory metrics alone would have reported this cell as a comfortable 8 GB run. PyTorch’s expandable-segments allocator option, the usual remedy for fragmentation, is not supported on this platform, and garbage-collection thresholds did not prevent the spill in a preliminary test. Preallocating the cache for the full sequence (`cache_implementation="static"`, run eagerly) removes the growth: batch 64 reserves 10.7 GB, runs in 4.0 s, and delivers 2,044 tok/s, versus 1,268 tok/s at batch 32, the largest batch the growing cache could run. The static cache is 24% slower at batch 1, where attention runs over the full preallocated length, but 4% faster at batch 16 and 15% faster at batch 32.

4.5 Generation length

Figure 5 plots latency against generated token count for BF16 eager batch 1. Increasing generation length from 32 to 128 raised end-to-end latency by roughly $4\times$, while tokens-per-second stayed between 52 and 54 tok/s. Under this harness, wall-clock time tracks decode length more strongly than the modest prompt-length change (64 vs. 256).

4.6 Compilation

Figure 6 reports eager/compile latency ratios. The median ratio was 1.015 (mean 1.027), ranging from 0.921 to 1.218. Thus `torch.compile` was approximately neutral on average for this model and measurement protocol, with occasional gains or regressions depending on the cell.

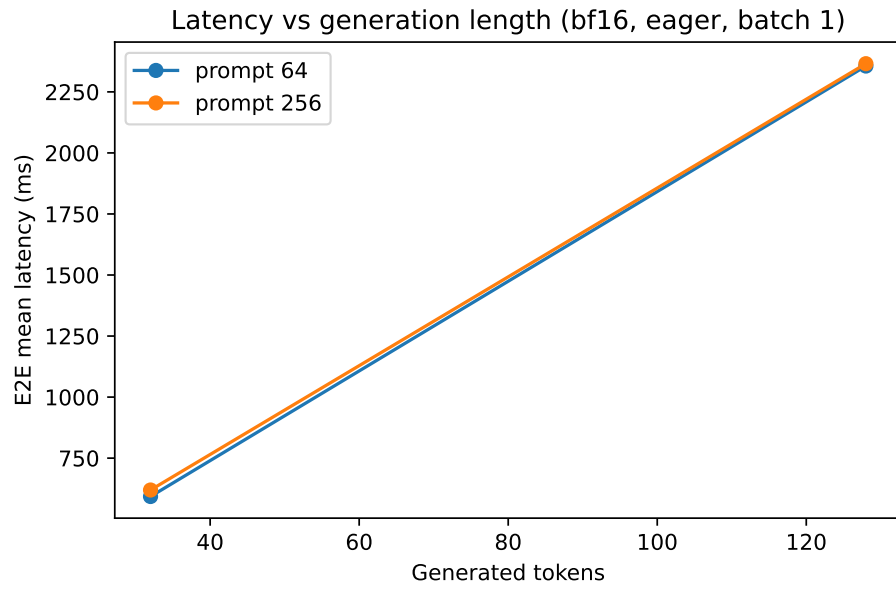


Figure 5: Latency versus generation length (BF16, eager, batch 1).

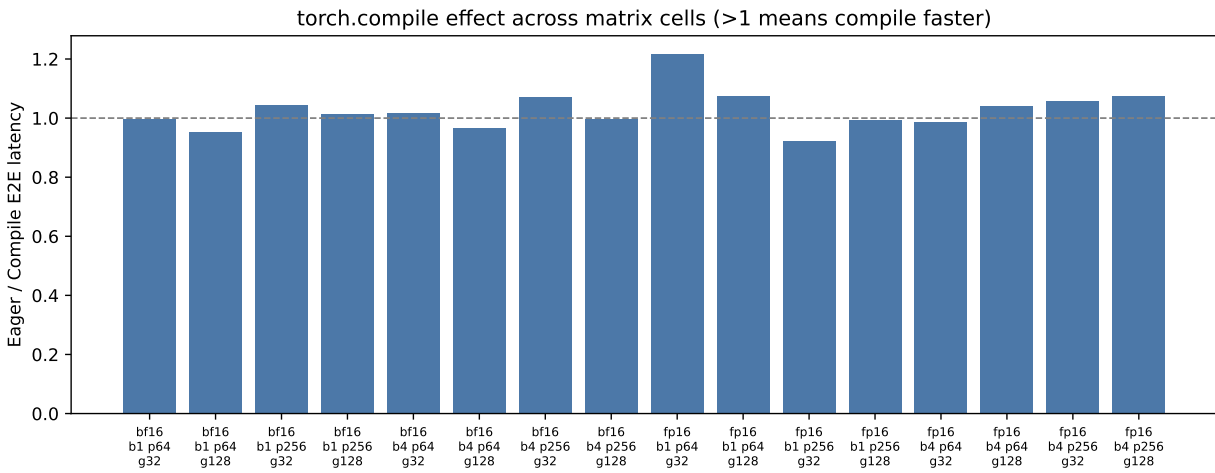


Figure 6: Eager/compile end-to-end latency ratios (> 1 means compile is faster).

5 Discussion and limitations

These measurements characterize one GPU, three models from a single family (with the factorial matrix on the smallest only), and an end-to-end `generate()` timer. They do not isolate time-to-first-token or inter-token latency, do not evaluate weight quantization or continuous batching under load, and do not claim optimality of any backend. The batch-size sweep uses a single configuration and reports its first, excluded pass openly; its two retained passes agreed closely, but results on a shared desktop workstation remain sensitive to host state. The model-size comparison confounds parameter count with architecture (depth, width, and attention type), which is itself one of its findings, and a second GPU would be needed to separate properties of the models from properties of the RTX 5070. Within that scope, the following conclusions are stable in the data: BF16 reduced latency vs. FP16; for the 135M model, batching up to 32 raised throughput with no latency cost, and beyond batch 32 latency grew roughly linearly while throughput saturated; the batch size at which batching stops being free falls as the model grows; small-batch step time tracks layer structure more than parameter count; a growing KV cache can inflate reserved memory several-fold and spill silently on Windows, which a static cache avoids; generation length dominated wall time; and `torch.compile` did not provide a consistent speedup.

6 Conclusion

We measured a full 32-cell inference matrix and a batch-size sweep on an RTX 5070 using a reproducible harness. The sweep shows a clear latency–throughput knee near batch 32 for the 135M model; the knee moves to batch 16 at 360M parameters and disappears at 1.7B, and for the largest model the choice of KV-cache implementation decides whether batch 64 fits on the device at all. The results emphasize workload- and precision-dependent behavior and caution against assuming that compilation alone improves short, small-model inference runs. A natural next step is to profile a decode step below and above the knee and to test CUDA Graphs, which remove per-step launch overhead and would therefore flatten the low-batch regime if our interpretation is correct.

Code and data availability

The harness, configurations, analysis scripts, and aggregated results are available at <https://github.com/benkarlsberg/GPU-Optimized-Transformer-Inference>.

References

- [1] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, volume 30, pages 5998–6008, 2017.
- [2] A. Paszke, S. Gross, F. Massa, A. Lerer, J. Bradbury, G. Chanan, T. Killeen, Z. Lin, N. Gimelshein, L. Antiga, A. Desmaison, A. Kopf, E. Yang, Z. DeVito, M. Raison, A. Tejani, S. Chilamkurthy, B. Steiner, L. Fang, J. Bai, and S. Chintala. PyTorch: An imperative style, high-performance deep learning library. In *Advances in Neural Information Processing Systems*, volume 32, pages 8024–8035, 2019.
- [3] A. Agrawal, N. Kedia, A. Panwar, J. Mohan, N. Kwatra, B. S. Gulavani, A. Tumanov, and R. Ramjee. Taming throughput-latency tradeoff in LLM inference with Sarathi-Serve. In *18th USENIX Symposium on Operating Systems Design and Implementation (OSDI)*, 2024.

- [4] L. Ben Allal, A. Lozhkov, E. Bakouch, L. von Werra, and T. Wolf. SmolLM: Blazingly fast and remarkably powerful. Hugging Face blog, <https://huggingface.co/blog/smollm>, 2024.
- [5] R. Y. Aminabadi, S. Rajbhandari, A. A. Awan, C. Li, D. Li, E. Zheng, O. Ruwase, S. Smith, M. Zhang, J. Rasley, and Y. He. DeepSpeed-Inference: Enabling efficient inference of transformer models at unprecedented scale. In *SC22: International Conference for High Performance Computing, Networking, Storage and Analysis*, 2022.
- [6] J. Ansel et al. PyTorch 2: Faster machine learning through dynamic Python bytecode transformation and graph compilation. In *Proceedings of the 29th ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)*, 2024.
- [7] T. Dao. FlashAttention-2: Faster attention with better parallelism and work partitioning. In *International Conference on Learning Representations (ICLR)*, 2024.
- [8] T. Dao, D. Y. Fu, S. Ermon, A. Rudra, and C. Ré. FlashAttention: Fast and memory-efficient exact attention with IO-awareness. In *Advances in Neural Information Processing Systems*, volume 35, 2022.
- [9] T. Dettmers, M. Lewis, Y. Belkada, and L. Zettlemoyer. LLM.int8(): 8-bit matrix multiplication for transformers at scale. In *Advances in Neural Information Processing Systems*, volume 35, 2022.
- [10] E. Frantar, S. Ashkboos, T. Hoeffer, and D. Alistarh. GPTQ: Accurate post-training quantization for generative pre-trained transformers. In *International Conference on Learning Representations (ICLR)*, 2023.
- [11] D. Kalamkar et al. A study of BFLOAT16 for deep learning training. arXiv:1905.12322, 2019.
- [12] W. Kwon, Z. Li, S. Zhuang, Y. Sheng, L. Zheng, C. H. Yu, J. E. Gonzalez, H. Zhang, and I. Stoica. Efficient memory management for large language model serving with PagedAttention. In *Proceedings of the 29th Symposium on Operating Systems Principles (SOSP)*, 2023.
- [13] P. Micikevicius, S. Narang, J. Alben, G. Diamos, E. Elsen, D. Garcia, B. Ginsburg, M. Houston, O. Kuchaiev, G. Venkatesh, and H. Wu. Mixed precision training. In *International Conference on Learning Representations (ICLR)*, 2018.
- [14] R. Pope, S. Douglas, A. Chowdhery, J. Devlin, J. Bradbury, J. Heek, K. Xiao, S. Agrawal, and J. Dean. Efficiently scaling transformer inference. In *Proceedings of Machine Learning and Systems (MLSys)*, volume 5, 2023.
- [15] V. J. Reddi et al. MLPerf inference benchmark. In *ACM/IEEE 47th Annual International Symposium on Computer Architecture (ISCA)*, 2020.
- [16] N. Shazeer. Fast transformer decoding: One write-head is all you need. arXiv:1911.02150, 2019.
- [17] Y. Sheng, L. Zheng, B. Yuan, Z. Li, M. Ryabinin, B. Chen, P. Liang, C. Ré, I. Stoica, and C. Zhang. FlexGen: High-throughput generative inference of large language models with a single GPU. In *Proceedings of the 40th International Conference on Machine Learning (ICML)*, 2023.
- [18] S. Williams, A. Waterman, and D. Patterson. Roofline: An insightful visual performance model for multicore architectures. *Communications of the ACM*, 52(4):65–76, 2009.
- [19] T. Wolf et al. Transformers: State-of-the-art natural language processing. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pages 38–45, 2020.

- [20] G.-I. Yu, J. S. Jeong, G.-W. Kim, S. Kim, and B.-G. Chun. Orca: A distributed serving system for transformer-based generative models. In *16th USENIX Symposium on Operating Systems Design and Implementation (OSDI)*, 2022.